

Homework 4

Problem 1

1. Hinge Loss is a loss function that's mainly used for binary classification, which is especially seen in support vector machines. Its purpose is to separate two classes while also maximizing the margin between them. If a prediction is correct as well as far from the decision boundary the hinge loss becomes 0 which then means no penalty is given. But if it's wrong or too close it'll give a penalty. Which makes it useful since it's pushed to be correct.

Source -

<https://www.geeksforgeeks.org/machine-learning/hinge-loss-relationship-with-support-vector-machines/>

2. On paper
3. On paper
4. Prediction rule : $\text{sign}(h(x_1, x_2))$

Predict:

- +1 if $h \geq 0$
- -1 otherwise

With final weights:

$$h(x_1, x_2) = -0.1 - 0.1x_1 + 0.4x_2$$

Results:

3 correct, 3 incorrect

Misclassification rate:

$$3/6 = 0.5$$

Final Answer: 50%

Problem 1

Question 2:

$$C(\hat{y}, y) = \begin{cases} 0 & \text{if } y\hat{y} \geq 1 \\ 1 - y\hat{y} & \text{if } y\hat{y} < 1 \end{cases}$$

$$y = w_0 + w_1 x_1 + w_2 x_2$$

$$C = 1 - y(w_0 + w_1 x_1 + w_2 x_2)$$

$$a) \frac{\partial C}{\partial w_0} = \begin{cases} 0 & \text{if } y\hat{y} \geq 1 \\ -y & \text{if } y\hat{y} < 1 \end{cases}$$

$$b) \frac{\partial C}{\partial w_1} = \begin{cases} 0 & \text{if } y\hat{y} \geq 1 \\ -yx_1 & \text{if } y\hat{y} < 1 \end{cases}$$

$$c) \frac{\partial C}{\partial w_2} = \begin{cases} 0 & \text{if } y\hat{y} \geq 1 \\ -yx_2 & \text{if } y\hat{y} < 1 \end{cases}$$

Question 3:

$$w_j \leftarrow w_j - \eta \frac{\partial C}{\partial w_j}$$

$$\text{Learning rate} = \eta = 0.1$$

$$\text{initial weights } w_0 = w_1 = w_2 = 0$$

$$P1 = (-4, 0, -1)$$

$$Pred = \hat{y} = 0$$

$$\text{Margin} = y\hat{y} = (-1)(0) = 0 < 1$$

$$\text{Subgradient } \frac{\partial C}{\partial w_0} = 1$$

$$\frac{\partial C}{\partial w_1} = -4$$

$$\frac{\partial C}{\partial w_2} = 0$$

$$\text{update } w_0 = 0 - 0.1(1) = -0.1$$

$$w_1 = 0 - 0.1(-4) = 0.4$$

$$w_2 = 0$$

$$\text{New weights} = (-0.1, 0.4, 0)$$

IT2:

$$(-1, 1, 1)$$

$$y = -0.1 + 0.4(1) = 0.3$$

$$(1)(0.3) = 0.3 < 1$$

$$\frac{\partial C}{\partial w_0} = -1 \quad w_0 = -0.1 - 0.1(-1) = 0$$

$$\frac{\partial C}{\partial w_1} = 1 \quad w_1 = 0.4 - 0.1(1) = 0.3$$

$$\frac{\partial C}{\partial w_2} = -1 \quad w_2 = 0 - 0.1(-1) = 0.1$$

$$\Rightarrow (0, 0.3, 0.1)$$

IT3:

$$(0, -1, -1)$$

$$y = -0.1$$

$$(-1)(-0.1) = 0.1 < 1$$

$$\frac{\partial C}{\partial w_0} = 1 \quad w_0 = 0 - 0.1(1) = -0.1$$

$$\frac{\partial C}{\partial w_1} = 0 \quad w_1 = 0.3$$

$$\frac{\partial C}{\partial w_2} = -1 \quad w_2 = 0.1 - 0.1(-1) = 0.2$$

$$\Rightarrow (-0.1, 0.3, 0.2)$$

IT4

$$(2, 1, 1)$$

$$y = -0.1 + 0.6(2) + 0.2 = 0.7$$

$$0.7 < 1$$

$$\frac{\partial C}{\partial w_0} = -1 \quad w_0 = 0$$

$$\frac{\partial C}{\partial w_1} = -2 \quad w_1 = 0.3$$

$$\frac{\partial C}{\partial w_2} = -1 \quad w_2 = 0.2$$

$$\Rightarrow (0, 0.5, 0.3)$$

IT5

$$(3, 0, 1)$$

$$\hat{y} = 1.5$$

$$1.5 > 1$$

$$\text{No update} \Rightarrow (0, 0.5, 0.3)$$

IT6

$$(6, -1, -1)$$

$$y = 3 - 0.3 = 2.7$$

$$(-1)(2.7) = -2.7 < 1$$

$$\frac{\partial C}{\partial w_0} = 1, w_0 = 0.1$$

$$\frac{\partial C}{\partial w_1} = 6, w_1 = -0.1$$

$$\frac{\partial C}{\partial w_2} = -1, w_2 = 0.2$$

$$\Rightarrow (-0.1, -0.1, 0.2)$$

Problem 2

$$1. -2(y_1 - (w_0 + w_1 x_1)) - 2(y_2 - (w_0 + w_1 x_2))$$

$$= dC/dw_0$$

$$-2x_1(y_1 - (w_0 + w_1x_1)) - 2x_2(y_2 - (w_0 + w_1x_2))$$

$$= dC/dw_1$$

1) $\frac{d}{dw_0} C(w) \quad \frac{d}{dw_1} C(w)$

$\hat{y}_i = w_0 + w_1 x_i$

$C(w) = (y_1 - \hat{y}_1)^2 + (y_2 - \hat{y}_2)^2$

$C(w) = (y_1 - (w_0 + w_1 x_1))^2 + (y_2 - (w_0 + w_1 x_2))^2$

$\rightarrow \frac{d}{dw_0} (y_1 - (w_0 + w_1 x_1))^2 \quad \rightarrow \frac{d}{dw_0} (y_2 - (w_0 + w_1 x_2))^2$

$2(y_1 - (w_0 + w_1 x_1)) \quad -2(y_2 - (w_0 + w_1 x_2))$

$\frac{d}{dw_0} C(w) = -2(y_1 - (w_0 + w_1 x_1)) - 2(y_2 - (w_0 + w_1 x_2))$

$$2. -2(y_1 - (w_0 + w_1 x_1)) - 2(y_2 - (w_0 + w_1 x_2)) + 2\lambda w_0$$

$$= dC^*/dw_0$$

$$-2x_1(y_1 - (w_0 + w_1 x_1)) - 2x_2(y_2 - (w_0 + w_1 x_2)) + 2\lambda w_1$$

$$= dC \sim / dw_1$$

$$\begin{aligned}
 2.) \quad \tilde{C}(w) &= (y_1 - (w_0 + w_1 x_1))^2 - 2(y_2 - (w_0 + w_1 x_2)) \\
 \frac{d}{dw_0} \lambda (w_0^2 + w_1^2) &= \lambda (2w_0) \\
 &= 2\lambda w_0 \\
 \frac{d}{dw_0} \tilde{C}(w) &= -2(y_1 - (w_0 + w_1 x_1)) - 2(y_2 - (w_0 + w_1 x_2)) + 2w_0 \\
 &= -2x_1(y_1 - (w_0 + w_1 x_1)) - 2x_2(y_2 - (w_0 + w_1 x_2)) \\
 \frac{d}{dw_1} \lambda (w_0^2 + w_1^2) &= 2\lambda w_1 \\
 \frac{d}{dw_1} \tilde{C}(w) &= -2x_1(y_1 - (w_0 + w_1 x_1)) - 2x_2(y_2 - (w_0 + w_1 x_2)) + 2\lambda w_1
 \end{aligned}$$

5. Average test error without regularization: about 1.8 to 2.0

Average test error with regularization: about 0.44

Ai Usage

Used Ai for general usage, term explanation, as well as double checking that we are on the correct path / using the proper techniques correctly.